# 1 AIGym

## 1.1 System overview

Native Rust development profile for local Java source `AI-gym/`. One independently configured executable, identity, port and state file per role. No Java, Go or Python process implements the service. Some essential operations are explicitly unavailable.

## 1.2 Design and trust boundaries

Tokio/Hyper/Axum with rustls enforces mutual TLS. Certificate chain, validity, local CN/profile and startup-configured optional CRLs are checked. CRLs need process restart to refresh; without CRLs no revocation enforcement is claimed. Tokens are verified with Authentication and bound to the TLS identity. Business calls require ConsumerAuthorization; management requires a verified sysop or explicit broad whitelist. CA provisioning instead applies Java-style named certificate-principal rights. Outbound authorities are pinned to expected CN/profile before credentials are sent, in addition to chain, SAN and time validation.

## 1.3 Persistent state

Independent locked atomic JSON replaces SQL. Clone/validate/fsync/rename commits prevent partially published local transactions. Rename is the commit point; directory-sync failure is an uncertain durable commit, not a rollback. Owner-only keys/password files are separate. Network effects are not distributed transactions; interrupted starts and bridge operations require reconciliation. No automatic Java state import, clustering or deployment is delivered.

## 1.4 Implemented behavior

Persisted samples, kNN regression, observation history

## 1.5 Open release gates

Broker publishing; local extension only. All systems remain development profiles, not certified interchangeable Java replacements. See `COMPATIBILITY.md` for detailed DTO fields and restrictions; the Java sources remain the authoritative contract.

## 1.6 Produced and consumed services

Advertised implemented HTTPS definitions: `aiGym`, `echo`, `generalManagement`.

Authorities consumed as applicable: Authentication, ServiceRegistry, ConsumerAuthorization and explicitly configured pinned adapter/provider origins. Only implemented operations are registered. Service versions are 1.0.0 except Gatekeeper/GateTunnel 5.2.2. No AMQP sysMonitor.observe implementation is advertised.

## 1.7 HTTPS operations

JSON unless raw configuration bytes, PEM CRL or echo text. Bodies: 1 MiB; remote responses: 8 MiB. Header: `Bearer IDENTITY-TOKEN//token`. Unknown routes return 404, absent dependencies 503. Unported operations return 501 after applicable access checks and are excluded from registration. Route presence is not semantic parity.

| Method | Path | Access / status |
|---|---|---|
| GET | /aigym/general/mgmt/get-config | management |
| POST | /aigym/general/mgmt/logs | management |
| GET | /aigym/monitor/echo | certificate |
| GET | /aigym/status | identity |
| POST | /aigym/samples | identity |
| POST | /aigym/predict | identity |
| GET | /aigym/observations | identity |
| GET | /health | certificate |

Error envelope: errorMessage, errorCode, exceptionType, origin and timestamp. Full Java status, DTO and validation equivalence requires differential testing.

## 2 References and qualification

See contracts.json for local Java SHA256 hashes/mappings, COMPATIBILITY.md for DTO details and gaps, and VERIFICATION.md for tests actually run. No upstream release-tag provenance, complete replacement or real-cloud interoperability is asserted.